

Bluestock Fintech Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline, Analytics and Dashboard Assets

Executive Summary

This project consolidates publicly available and supplied mutual fund datasets into a reproducible analytics platform. It includes data ingestion, cleaning, SQLite storage, exploratory analysis, risk-return metrics, advanced investor analytics, dashboard-ready extracts, and final submission documents.

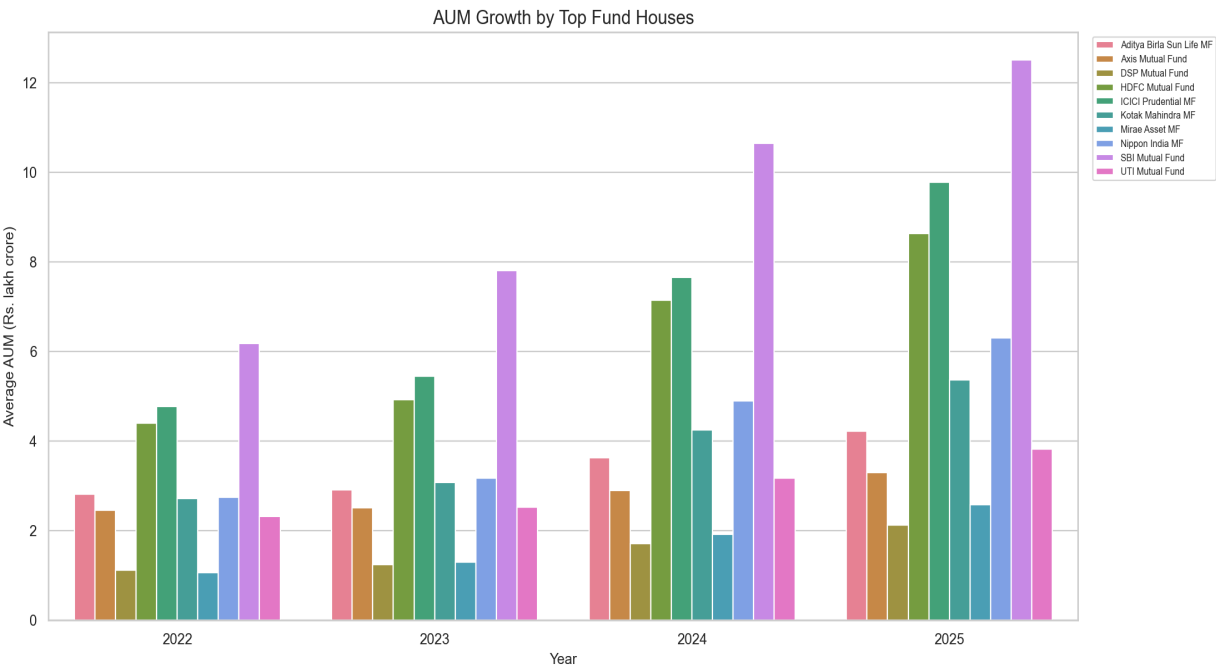
Dataset Coverage

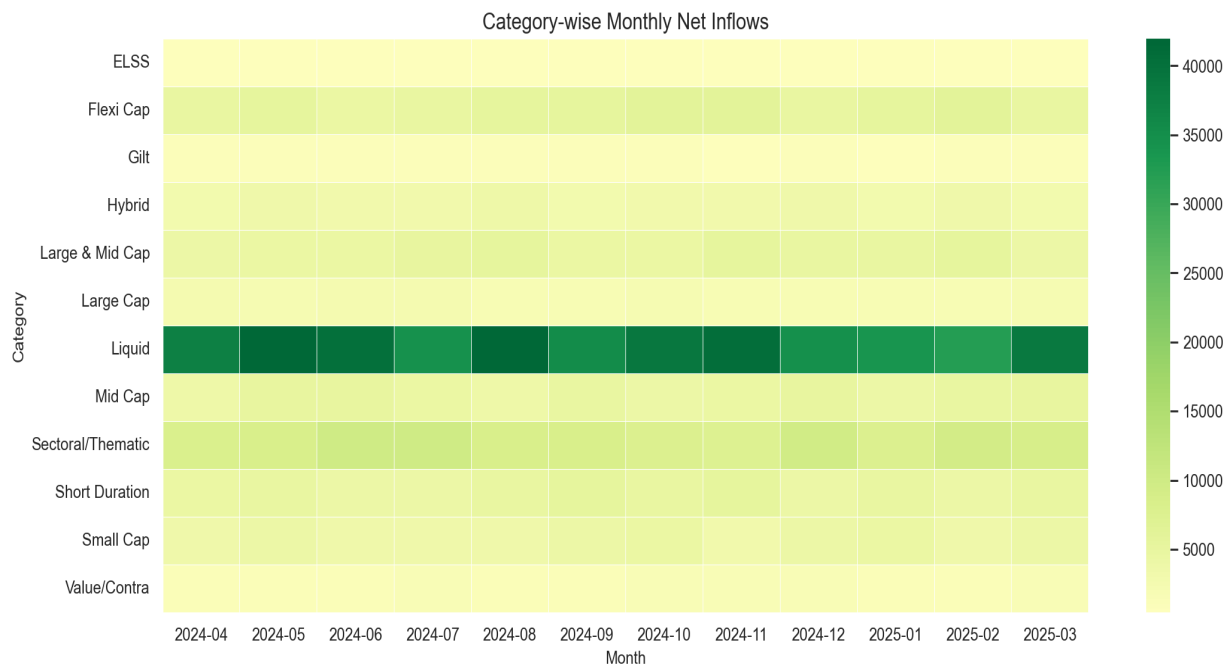
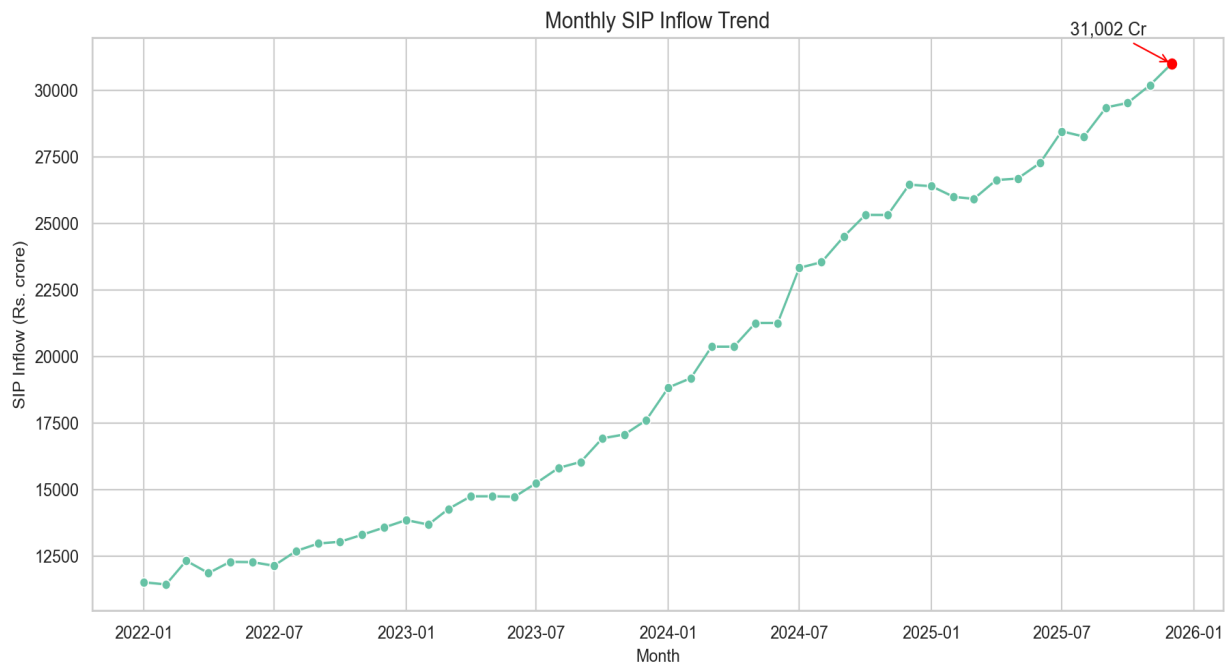
The project covers 40 schemes, 46,000 NAV observations, 32,778 investor transactions, benchmark index series, AUM, SIP, folio, category inflow, and portfolio holdings datasets.

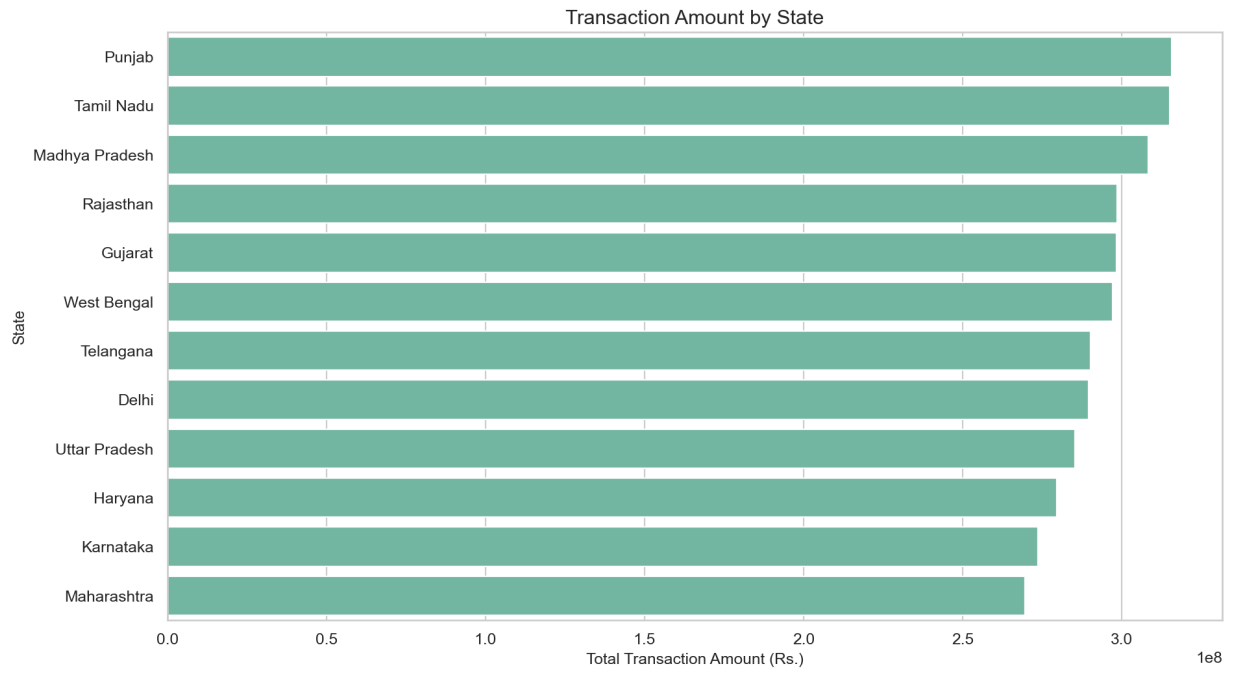
System Architecture

The pipeline follows Extract, Transform, Load, Analyse, and Visualise layers. Raw CSVs are loaded with Pandas, cleaned into processed outputs, stored in SQLite tables, analysed with Python scripts/notebooks, and exported for Power BI/Tableau usage.

EDA Highlights



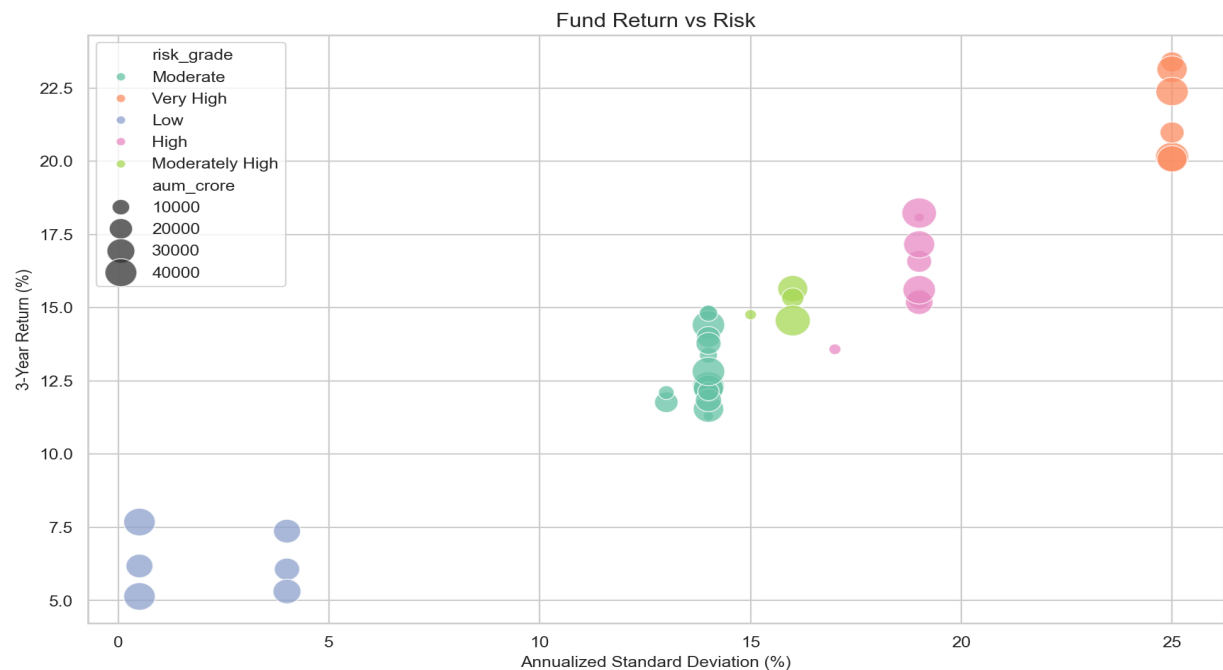
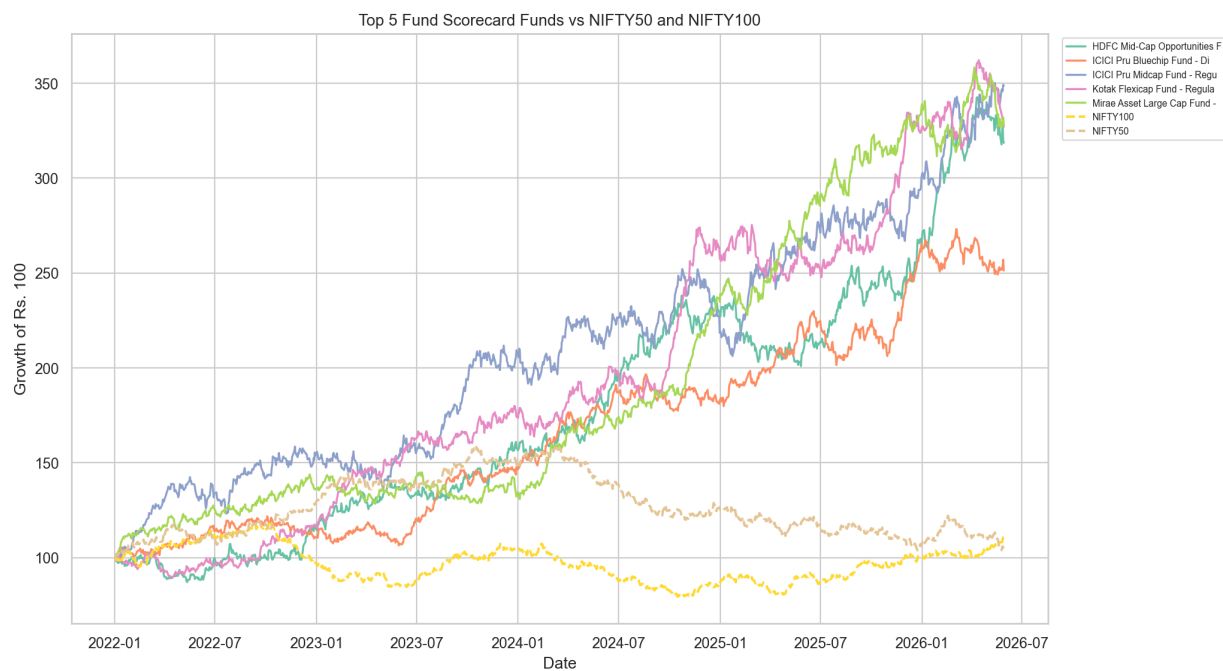




Performance Analytics

Day 4 computes annualized returns, 1/3/5-year CAGR, Sharpe, Sortino, Alpha, Beta, tracking error, information ratio, maximum drawdown, annual volatility, and a weighted composite scorecard.

fund_name	fund_house	return_3yr_pct	sharpe_ratio	alpha_pct	beta	fund_score
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	33.991970400445304	1.4472544182205025	26.96028200453649	0.0236852840158849	86.25
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	31.769242852237323	1.1792906349000456	29.238130993574853	0.000550876013726	82.25
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	29.57511058901836	1.3058136233276143	27.30669404266793	-0.0228278216758267	82.0
HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	32.43397122906222	1.0929210290197424	27.171701019636885	0.0051052653051613	80.75
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	32.478927222623	1.0256807136488637	21.176392561723205	0.0162335254291263	80.0
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	35.10252788903541	0.9975028263019924	26.053988588287343	-0.0662637607069823	77.0
SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	30.448613060027817	1.2073241814126952	23.180827809890687	-0.0317499571515098	74.81
Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	29.17135767741186	1.2340680357894307	28.245779365223942	0.0181356923187083	73.69



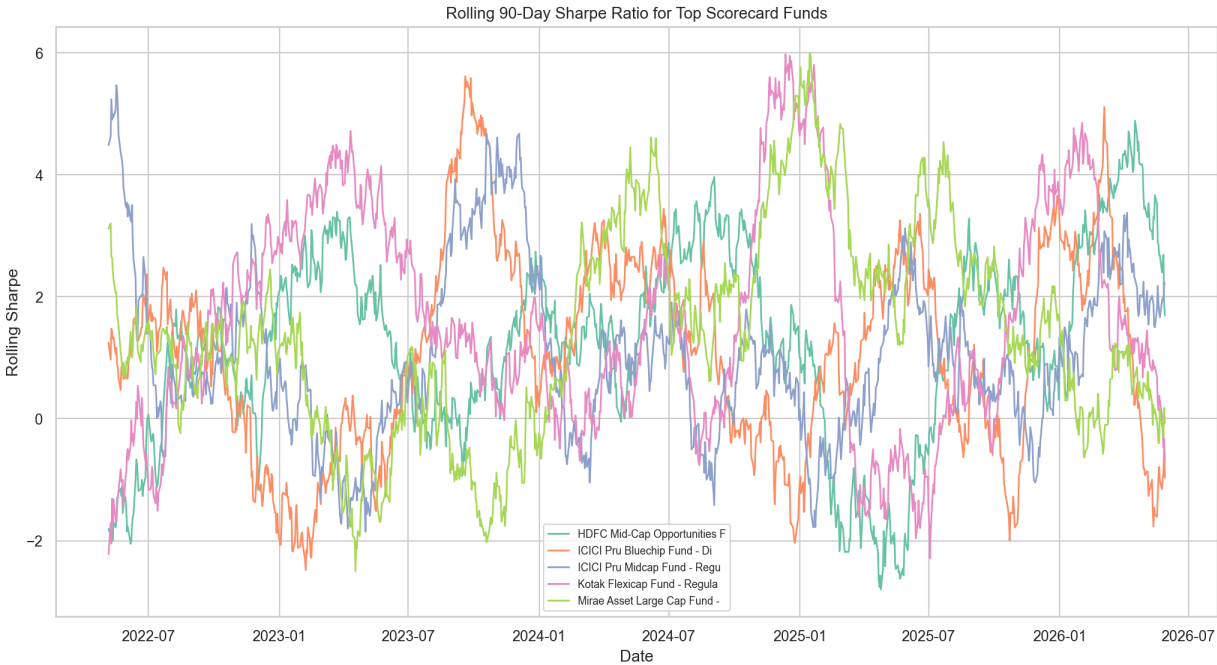
Advanced Analytics

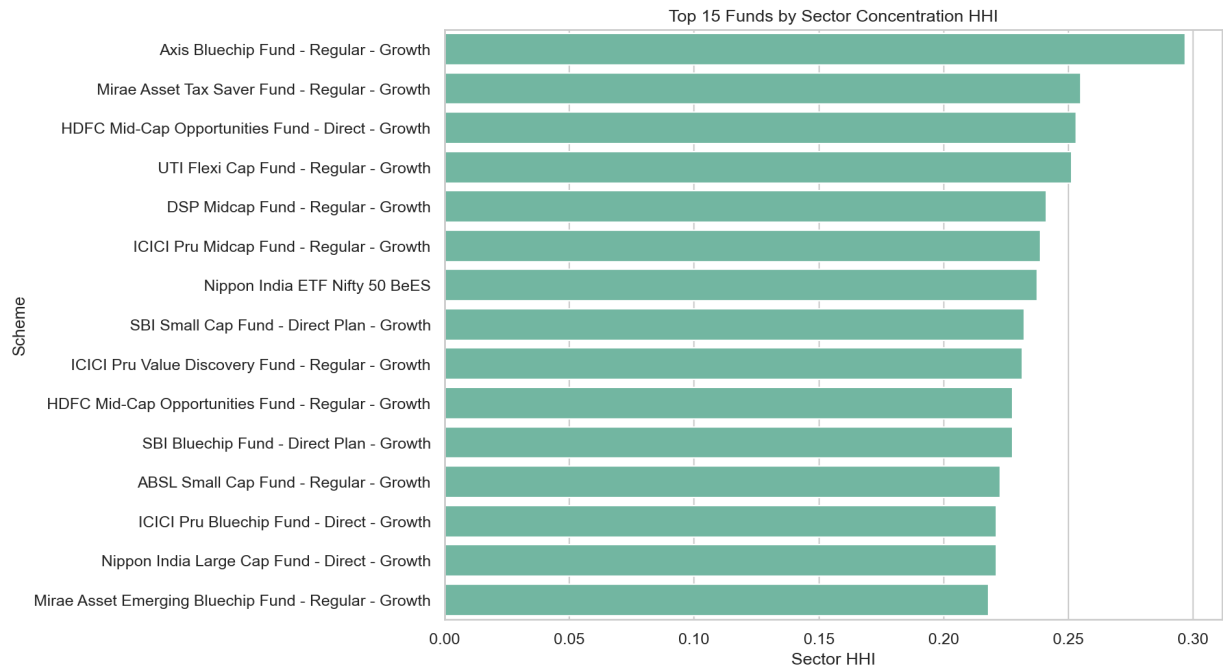
Day 6 adds historical VaR/CVaR, rolling Sharpe, cohort analysis, SIP continuity checks, recommendation logic, and sector concentration HHI.

scheme_name	fund_house	var_95_pct	cvar_95_pct
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	-2.6855452829253004	-3.238412245049218
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	-2.617777911363941	-3.166729360598028
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	-2.601987791067576	-3.245905898704435
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	-2.542848502985948	-3.2304072697818533
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	-2.4506409437533185	-3.0595264108802414
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	-2.34763680730338	-3.103625202188441

cohort_year	investors	avg_transaction_amount	total_invested	sip_transactions
2024	4803	107422.54183205638	3491125187	19549
2025	197	109158.5770609319	30455243	167

scheme_name	fund_house	sector_hhi
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund	0.29676909
Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	0.25499194
HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	0.253155
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	0.25138255
DSP Midcap Fund - Regular - Growth	DSP Mutual Fund	0.24107664
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	0.23869504





Dashboard Plan

Dashboard-ready extracts are available in dashboard/extracts. The Power BI build guide defines four report pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.

Recommendations

Use the fund scorecard for shortlist creation, review downside risk with VaR and max drawdown, and combine SIP/cohort insights with geographic segmentation for investor engagement. Direct plans show lower expense ratios and should be compared carefully against regular plans on risk-adjusted returns.

Limitations

Investor transaction data is synthetic, portfolio holdings are point-in-time, and dashboard construction in Power BI requires manual layout in Power BI Desktop. This analysis is educational and not financial advice.

Submission Checklist

ETL scripts, cleaned CSVs, SQLite schema/database, SQL queries, EDA charts, performance metrics, advanced analytics, dashboard extracts, final PDF report, PowerPoint deck, README, and local Git commits are complete.